Forty-seven squares: the prime-side / zero-side identity on V

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September 1, 2026 — lemma campaign, item 4

Abstract

Journal step 81 named the next lock-picking stroke: certify that the 47×47 matrix of the truncated Weil form at $\mu = 11$, assembled from primes, agrees with the Gram matrix of the zero-evaluators $\eta \mapsto \hat{\eta}(\gamma)$. MUSIC already saw the identity as a fact of subspaces. This note records the exact pairing, ships a replay script against `zeros280.pkl`, and states what ball arithmetic still owes.

1 The identity

On $V = \mathrm{span}(\eta_0, \dots, \eta_{46})$ at $L = \log 11$, the explicit formula says that the truncated Weil form assembled from the pole, the archimedean place and the prime towers equals, term by term,

$$Q(\eta_n, \eta_m) \;=\; \sum_{\gamma} |\widehat{\eta_n}(\gamma)|\,|\widehat{\eta_m}(\gamma)| \cdot \mathrm{sign}_{nm}(\gamma) \;=\; \sum_{\gamma} \widehat{\eta_n}(\gamma)\,\widehat{\eta_m}(\gamma),$$

the γ running over the nontrivial zeros of ζ (even window: γ and $-\gamma$ identified). The Fourier transforms of the shifted-cosine basis are closed form:

$$\widehat{\eta_0}(\gamma) = \frac{2\sin(\gamma L/2)}{\gamma\sqrt{L}}, \qquad \widehat{\eta_k}(\gamma) = \sqrt{\frac{2}{L}}\,\frac{2\gamma\sin(\gamma L/2)}{\gamma^2 - \omega_k^2} \quad (k \geq 1).$$

Call Q^{pr} the matrix built from $P + A - \sum_p T_p$ and Q^{z} the matrix with entries $\sum_\gamma \widehat{\eta_n}(\gamma)\widehat{\eta_m}(\gamma)$. The 47 squares are the diagonal of Q^{z}; the theorem wanted is $Q^{\mathrm{pr}} = Q^{\mathrm{z}}$ as matrices on V, with certified enclosures.

This is not RH. It is the explicit formula restricted to a finite-dimensional even window, with the usual archimedean normalization already checked on η_0 by `weil_normalization_check.py`.

2 What is already seen

MUSIC [?] §16.1 confirmed the identity as a subspace fact: the prime-assembled matrix and the zero-assembled matrix share the same radical, recover γ_1 to 10^{-19}, and drop at the band frontier $\omega_{\max} \approx 120.5$. That is a measurement of $Q^{\mathrm{pr}} \approx Q^{\mathrm{z}}$ on the quasi-null flag. It does not enclose every entry.

The public normalization check certifies two scalar identities (pole exact, archimedean against a ψ-integral to the $1/T$ tail) on the single vector η_0. Forty-six more diagonal squares, and the off-diagonal Gram, remain.

3 The script

`code/squares47.py` rebuilds Q^{pr} (mpmath, same closed Θ as the quorum engine) and Q^{z} from `zeros280.pkl`, and prints

$$\max_{n,m \leq N_0} \left| Q^{\mathrm{pr}}_{nm} - Q^{\mathrm{z}}_{nm} \right| \quad \text{and the relative residual on the diagonal.}$$

Default $N_0 = 6$ (cheap). $N_0 = 46$ is the full 47 squares and takes longer. The script does not use flint; promoting the same difference to Arb balls is the certification stroke, identical in kind to `positivite_certifiee.py`.

4 What certification must enclose

Three errors, all explicit:

1. quadrature / closed-form evaluation of P, A, T_p (already ball-controlled in the positivity script, radii $\leq 10^{-55}$);
2. tail of $\sum_{\gamma > \gamma_{\text{cut}}} \widehat{\eta_n}(\gamma)\widehat{\eta_m}(\gamma)$, algebraic $O(1/\gamma)$ as measured by MUSIC, hence a bound $O(\gamma_{\text{cut}}^{-1})$ after summing;
3. conversion of the known zeros to Arb (mpmath `zetazero` at working *dps*, or a frozen dyadic file).

Item 2 is the only new estimate. A cut at $\gamma = 280$ (the shipped cache) is enough for a 10^{-6} residual on the bulk of V and should be enough to see the razor 3.58×10^{-48} on the ground state only after the tail is budgeted, not ignored.

5 Status

The identity is the explicit formula on V. MUSIC plus the η_0 normalization check make it a measurement. The 47 certified squares are the same identity with radii. The script is the replay; flint is the certificate.

References

[1] D. Joubert, *Le milieu des nombres premiers*, §15.6–16.3.

Addendum — the tail and the endpoint, 1 September afternoon

Notebook §16.4, §16.7. In-band zeros carry 42% of λ_0; the rest is the out-of-band trail. The endpoint law $|\hat{v}_0(\gamma_1)| \approx C\lambda_0$ says the first square on the ground state is $C^2\lambda_0^2$, not λ_0 — a certificate of the 47 squares must resolve that λ^2 slot, which at $\mu = 16$ sits at 10^{-144}. Use `zeros_zeta_90_hp.pkl` and load it after setting *dps*.

Addendum — replay, 1 September

`code/squares47.py` now loads `zeros_zeta_90_hp.pkl` after setting *dps*, then appends the $\gamma > 133$ tail from `zeros280.pkl`. At $N_0 = 4$ (5-dimensional block): $\max |Q^{\text{pr}} - Q^{\text{z}}| = 6.3 \times 10^{-3}$, max relative diagonal residual 6.1%. The residual is the $\gamma > 280$ tail, not a *dps* artifact. Flint certification of the same difference is still the missing stroke.

Addendum — tail enclosed, 1 September

Extended cache `zeros500.pkl` ($\gamma_{500} = 811$). Replay at $N_0 = 4$: residual 4.1% (was 6.1% at $\gamma = 514$). Explicit envelope on the remainder

$$\Big|Q^{\text{z}}_{nm} - \sum_{|\gamma| \leq G} \hat{\eta}_n(\gamma)\hat{\eta}_m(\gamma)\Big| \;\leq\; 2c^2 \int_G^\infty \frac{\frac{1}{2\pi}\log(g/2\pi) + 1}{g^2}\, dg$$

with $c = 4/\sqrt{L}$, gives 3.2×10^{-2} at $G = 811$. Combined budget on each entry of the 5×5 block: $|Q^{\text{pr}} - Q^{\text{z}}| \leq 4.3 \times 10^{-3} + 3.2 \times 10^{-2}$. This is a remainder, not an Arb certificate of equality. The identity on V is measured at four percent plus an explicit $O(1/G)$ tail. Flint can take the same difference to balls; the zeros are not yet Arb objects.